Stony Brook Institute at Anhui University

安徽大学纽约石溪学院

Chapter 5
Basic Concepts of Statistics

5.3 Common sampling distribution

- ***Theorem 5.3.1***

If $X_1, X_2, \ldots, X_n$ is independent random variance. $X_i \sim N(\mu_i, \sigma_i^2)$. $c_1, c_2, \ldots, c_n$ are constants, then statistic $T = \sum_{i=1}^{n} c_i X_i$ has the result as follows:

$$T = \sum_{i=1}^{n} c_i X_i \sim N\left(\sum_{i=1}^{n} c_i \mu_i, \sum_{i=1}^{n} c_i^2 \sigma_i^2\right)$$

- ***Corollary 5.3.1***

If $X_1, X_2, \ldots, X_n$ is a random sample from a normal population $N(\mu, \sigma^2)$, $c_1, c_2, \ldots, c_n$ are constants, then

$$T = \sum_{i=1}^{n} c_i X_i \sim N\left(\mu \sum_{i=1}^{n} c_i, \sigma^2 \sum_{i=1}^{n} c_i^2\right).$$

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Theorem 5.3.2

If $X_1, X_2, \ldots, X_n$ is a random sample from a normal population $N(\mu, \sigma^2)$, the sample mean and sample variace follow:

a. $\overline{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$

b. $\frac{(n-1)S^2}{\sigma^2} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(X_i - \overline{X})^2 \sim \chi^2(n-1);$

c. $\overline{X}, S^2$ are independent.

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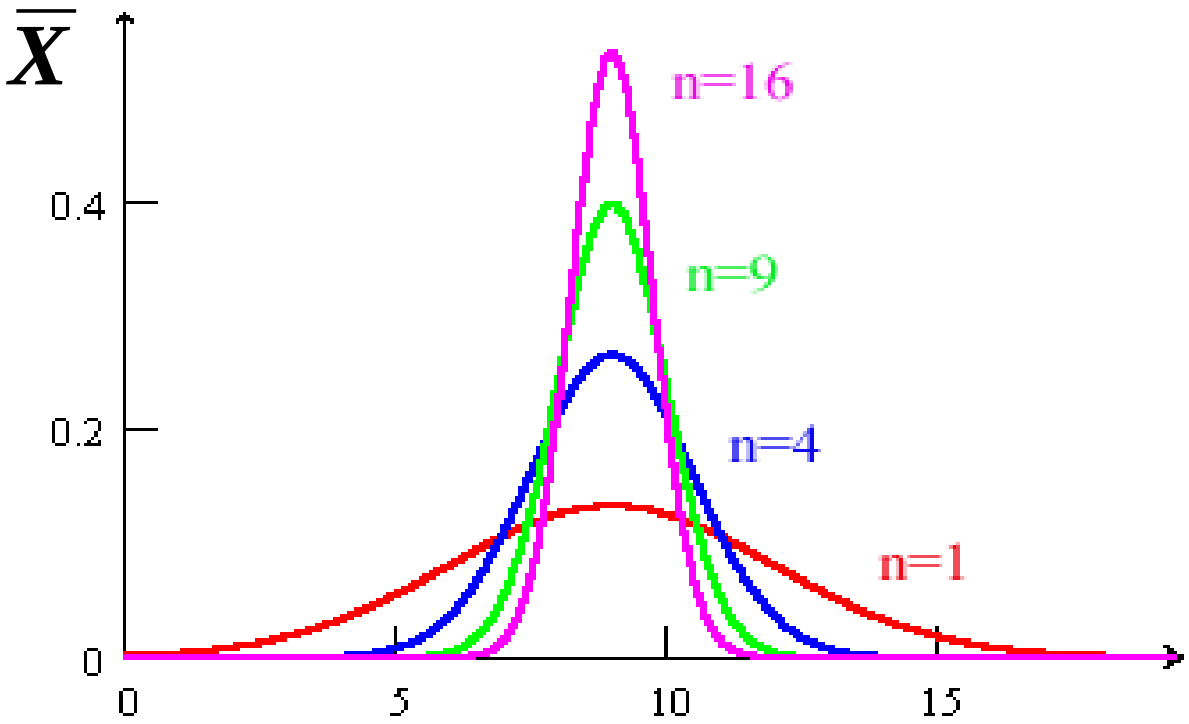

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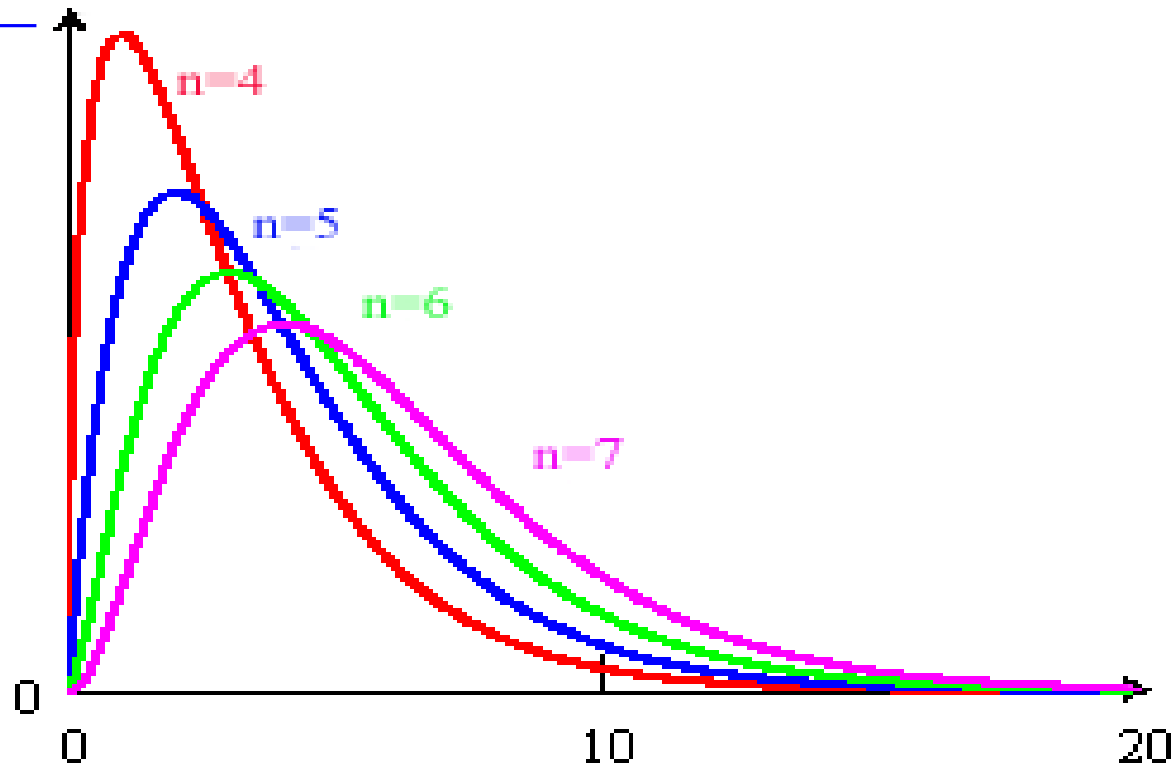
$$\frac{(n-1)S^2}{\sigma^2}$$



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Theorem 5.3.3

If $X_1, X_2, \ldots, X_n$ is a random sample from a normal population $N(\mu, \sigma^2)$, the sample mean $\bar{X}$ and sample variace S^2 follow:

$$\frac{\sqrt{n}(\bar{X}-\mu)}{S} \sim t(n-1).$$

Proof: From Theorem 5.3.2, $\frac{\sqrt{n}(\bar{X}-\mu)}{\sigma} \sim N(0,1)$, $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$,

$\bar{X}$ and S^2 *are independent, so by the definition of t distribution,*

$$\frac{\sqrt{n}(\bar{X}-\mu)}{S} = \frac{\sqrt{n}(\bar{X}-\mu)}{\sigma} \Bigg/ \sqrt{\frac{(n-1)S^2}{\sigma^2(n-1)}} \sim t(n-1)$$

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Theorem 5.3.4

If $X_1, X_2, \ldots, X_{n1}$ is a random sample from a normal population $N(\mu_1, \sigma_1^2)$, $Y_1, Y_2, \ldots, Y_{n2}$ is a random sample from another normal population $N(\mu_2, \sigma_2^2)$, the two samples are independent of each other. $\bar{X}$, $\bar{Y}$ are the two sample means of the two samples. We have

$$\bar{X} - \bar{Y} \sim N(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})$$

or

$$\frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{\sigma_1^2 / n_1 + \sigma_2^2 / n_2}} \sim N(0,1)$$

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Theorem 5.3.5

If $X_1, X_2, \ldots, X_{n1}$ is a random sample from a normal population $N(\mu_1, \sigma_1^2)$, $Y_1, Y_2, \ldots, Y_{n2}$ is a random sample from another normal population $N(\mu_2, \sigma_2^2)$, the two samples are independent of each other. $\bar{X}$, $\bar{Y}$ are the two sample means of the two samples, S_1^2, S_2^2 are the two sample variable, then

$$\text{(a)} \quad \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F(n_1 - 1, n_2 - 1);$$

$$\text{(b)} \quad if \ \sigma_1^2 = \sigma_2^2 = \sigma^2 ,$$

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$$\frac{(\overline{X}-\overline{Y})-(\mu_1-\mu_2)}{S_w\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} \sim t(n_1+n_2-2),$$

$$here,\ \ S_w^2=\frac{(n_1-1)S_1^2+(n_2-1)S_2^2}{n_1+n_2-2},\quad S_w=\sqrt{S_w^2}.$$

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Example 5.3.1:

If $X_1, X_2, \ldots, X_n$ is a random sample from a normal population $N(0, \sigma^2)$, what is the distribution of statistic

$$T = \frac{X_1 + X_2}{\sqrt{X_3^2 + X_4^2}} \quad ?$$

$$Solution: X_1 + X_2 \sim N(0, 2\sigma^2), \qquad so \ \frac{X_1 + X_2}{\sqrt{2\sigma^2}} \sim N(0,1)$$

$$\because \frac{X_3}{\sqrt{\sigma^2}}, \frac{X_4}{\sqrt{\sigma^2}} \quad i.i.d. \sim N(0,1),$$

$$\therefore \quad \frac{X_3^2}{\sigma^2} + \frac{X_4^2}{\sigma^2} \sim \chi^2(2)$$

from the definition of t distribution, we know

$$\frac{\dfrac{X_1 + X_2}{\sqrt{2\sigma^2}}}{\sqrt{\dfrac{X_3^2 + X_4^2}{\sigma^2 \cdot 2}}} \sim t(2) \quad \Rightarrow \quad \frac{X_1 + X_2}{\sqrt{X_3^2 + X_4^2}} \sim t(2)$$

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Example 5.3.2 If $X_1, X_2, \ldots, X_n, \ldots, X_{n+m}$ is a random sample from a normal population $N(0,\sigma^2)$. Find the probability distribution of statistic

$$T = \frac{\sqrt{m}\sum_{i=1}^{n} X_i}{\sqrt{n}\sqrt{\sum_{i=n+1}^{n+m} X_i^2}}$$

Solution:

Since $X_1, X_2, \ldots, X_n, \ldots, X_{n+m}$ are independent, $\frac{X_i}{\sigma} \sim N(0,1)$, so

$$\sum_{i=1}^{n}\left(\frac{X_i}{\sigma}\right) \sim N(0,n), \Rightarrow \quad \sum_{i=1}^{n}\frac{X_i}{\sqrt{n}\sigma} \sim N(0,1)$$

$$\frac{X_i}{\sigma} \sim N(0,1) \Rightarrow \quad \sum_{i=n+1}^{n+m}\left(\frac{X_i}{\sigma}\right)^2 \sim \chi^2(m)$$

Since $\sum_{i=1}^{n}(\frac{X_i}{\sqrt{n}\sigma})$ and $\sum_{i=n+1}^{n+m}(\frac{X_i}{\sigma})^2$ are independent,

so from the definition of t distribution, we know

$$\frac{\sum_{i=1}^{n}(X_i/\sqrt{n}\sigma)}{\sqrt{\sum_{i=n+1}^{n+m}(X_i/\sigma)^2/m}} \sim t(m)$$

$$\Rightarrow T = \frac{\sqrt{m}\sum_{i=1}^{n}X_i}{\sqrt{n}\sqrt{\sum_{i=n+1}^{n+m}X_i^2}} \sim t(m)$$

Example A sample with a capacity of 36 is randomly selected from population $N(52, 6.3^2)$, calculate the probability that $\overline{X}$ is between 50.8 and 53.8.

Solution : **n=36**

$$\overline{X} \sim N(52, \frac{6.3^2}{36}) = N(52, 1.05^2)$$

$$P(50.8 < \overline{X} < 53.8) = P\{\frac{50.8-52}{1.05} < \frac{\overline{X}-52}{1.05} < \frac{53.8-52}{1.05}\}$$

$$= \Phi(1.71) - \Phi(-1.14)$$

$$= \Phi(1.71) + \Phi(1.14) - 1$$

$$= 0.9564 + 0.8729 - 1 = 0.8293$$

Example. Known $X \sim t(n)$, Verify: $X^2 \sim F(1,n)$

Solution :

$$\because \ X \sim t(n),$$

$$X = \frac{U}{\chi(n)/\sqrt{n}} \ , U \sim N(0,1)$$

$$\therefore \ X^2 = \left(\frac{U}{\chi(n)/\sqrt{n}}\right)^2 = \frac{U^2}{\chi^2(n)/n} = \frac{\chi^2(1)/1}{\chi^2(n)/n}$$

$$\Rightarrow \quad X^2 \sim F(1,n)$$